

```
In [1]: import pandas as pd  
import seaborn as sns  
import matplotlib.pyplot as plt
```

```
In [3]: #Task 1 – Instagram Followers vs Posts  
  
df = pd.read_csv("instagram_users(1).csv")
```

```
In [4]: df
```

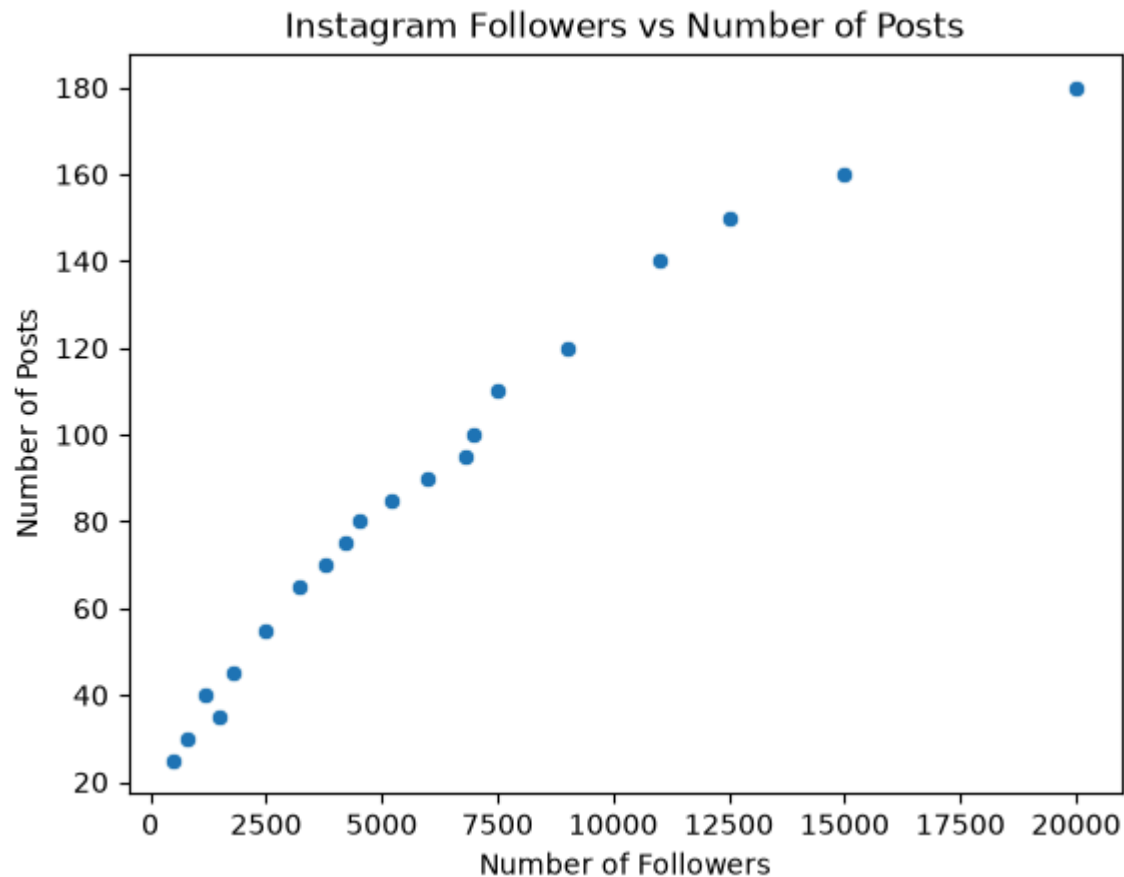
Out[4]:

	user	followers	posts
0	User_1	500	25
1	User_2	1200	40
2	User_3	2500	55
3	User_4	800	30
4	User_5	4500	80
5	User_6	3200	65
6	User_7	7000	100
7	User_8	1500	35
8	User_9	9000	120
9	User_10	6000	90
10	User_11	11000	140
11	User_12	3800	70
12	User_13	15000	160
13	User_14	5200	85
14	User_15	7500	110
15	User_16	1800	45
16	User_17	12500	150
17	User_18	4200	75
18	User_19	20000	180
19	User_20	6800	95

In [5]: `sns.scatterplot(x="followers", y="posts", data=df)`

```
plt.xlabel("Number of Followers")  
plt.ylabel("Number of Posts")  
plt.title("Instagram Followers vs Number of Posts")
```

Out[5]: Text(0.5, 1.0, 'Instagram Followers vs Number of Posts')



Conclusion The scatter plot shows the relationship between the number of followers and the number of posts for 20 Instagram users.

In [22]: *#Task 2 – Zomato Daily Orders*

```
df = pd.read_csv("zomato_daily_orders.csv")
```

In [23]: df

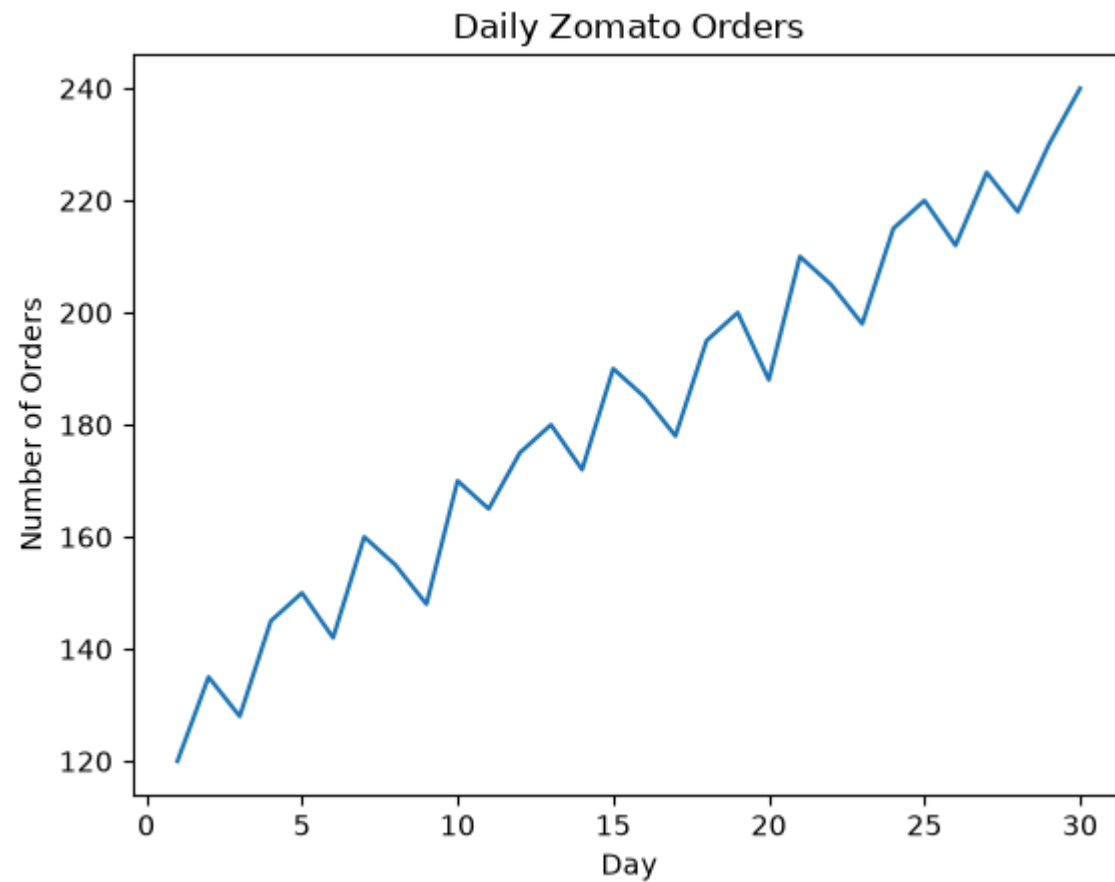
Out[23]:

	day	orders
0	1	120
1	2	135
2	3	128
3	4	145
4	5	150
5	6	142
6	7	160
7	8	155
8	9	148
9	10	170
10	11	165
11	12	175
12	13	180
13	14	172
14	15	190
15	16	185
16	17	178
17	18	195
18	19	200
19	20	188
20	21	210
21	22	205

	day	orders
22	23	198
23	24	215
24	25	220
25	26	212
26	27	225
27	28	218
28	29	230
29	30	240

```
In [26]: sns.lineplot(x="day", y="orders", data=df)
plt.title("Daily Zomato Orders")
plt.xlabel("Day")
plt.ylabel("Number of Orders")
```

```
Out[26]: Text(0, 0.5, 'Number of Orders')
```



#Conclusion The line plot shows the daily trend in Zomato food orders over one month. It helps us understand whether the number of orders is increasing or decreasing.

```
In [28]: #Task 3 – Movie Ratings vs Reviews
```

```
df = pd.read_csv("movies(1).csv")
```

```
In [29]: df
```

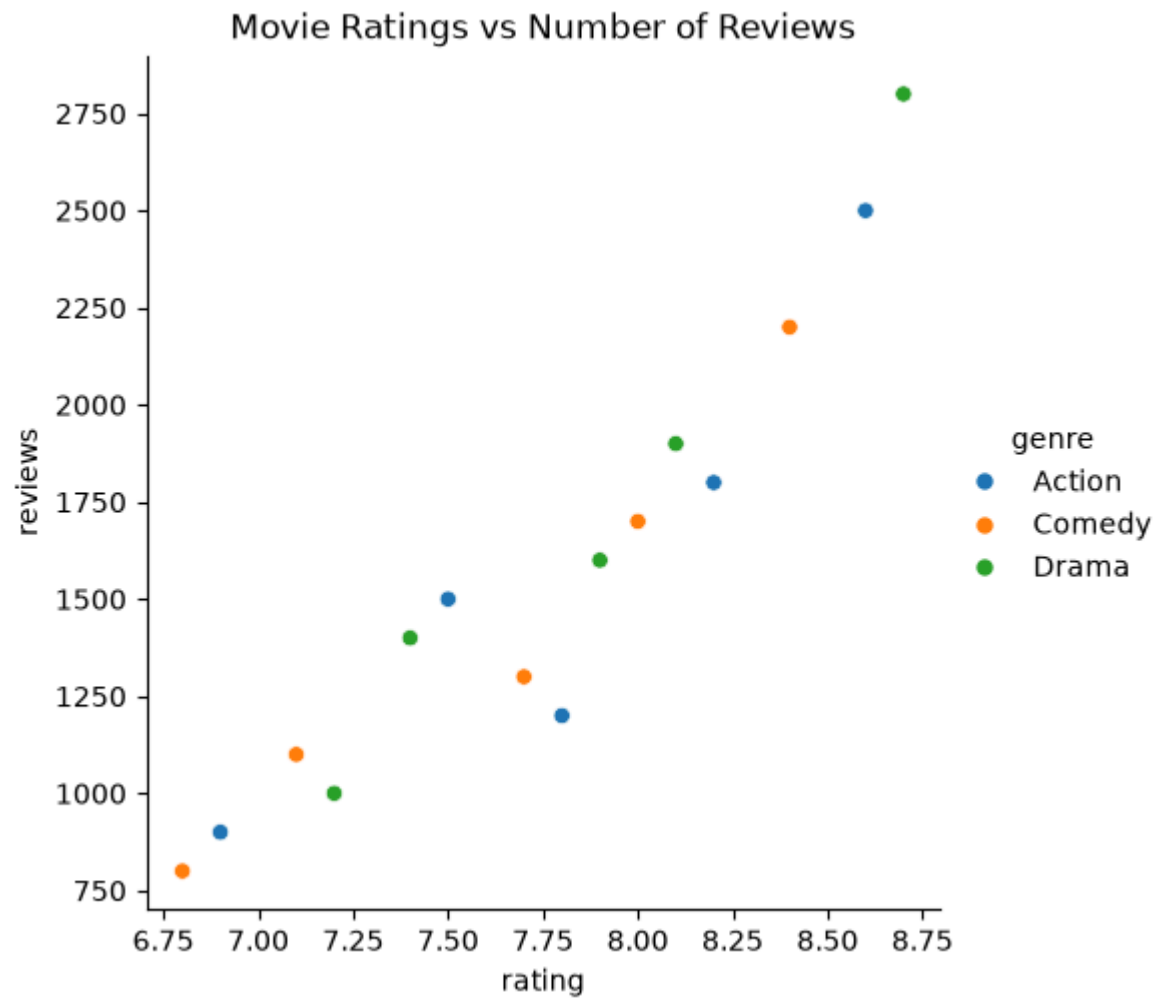
Out[29]:

	movie	rating	reviews	genre
0	Movie 1	7.8	1200	Action
1	Movie 2	8.2	1800	Action
2	Movie 3	6.9	900	Action
3	Movie 4	7.5	1500	Action
4	Movie 5	8.6	2500	Action
5	Movie 6	7.1	1100	Comedy
6	Movie 7	8.0	1700	Comedy
7	Movie 8	7.7	1300	Comedy
8	Movie 9	8.4	2200	Comedy
9	Movie 10	6.8	800	Comedy
10	Movie 11	7.9	1600	Drama
11	Movie 12	8.7	2800	Drama
12	Movie 13	7.2	1000	Drama
13	Movie 14	8.1	1900	Drama
14	Movie 15	7.4	1400	Drama

```
In [30]: sns.relplot(x="rating", y="reviews", hue="genre", data=df)

plt.title("Movie Ratings vs Number of Reviews")
```

```
Out[30]: Text(0.5, 1.0, 'Movie Ratings vs Number of Reviews')
```



#Conclusion The relplot shows the relationship between movie ratings and the number of reviews. The hue parameter separates the movies into Action, Comedy, and Drama genres.

```
In [31]: #Task 4 – YouTube Hours vs Exam Scores
```

```
df = pd.read_csv("youtube_exam_scores.csv")
```

```
In [32]: df
```

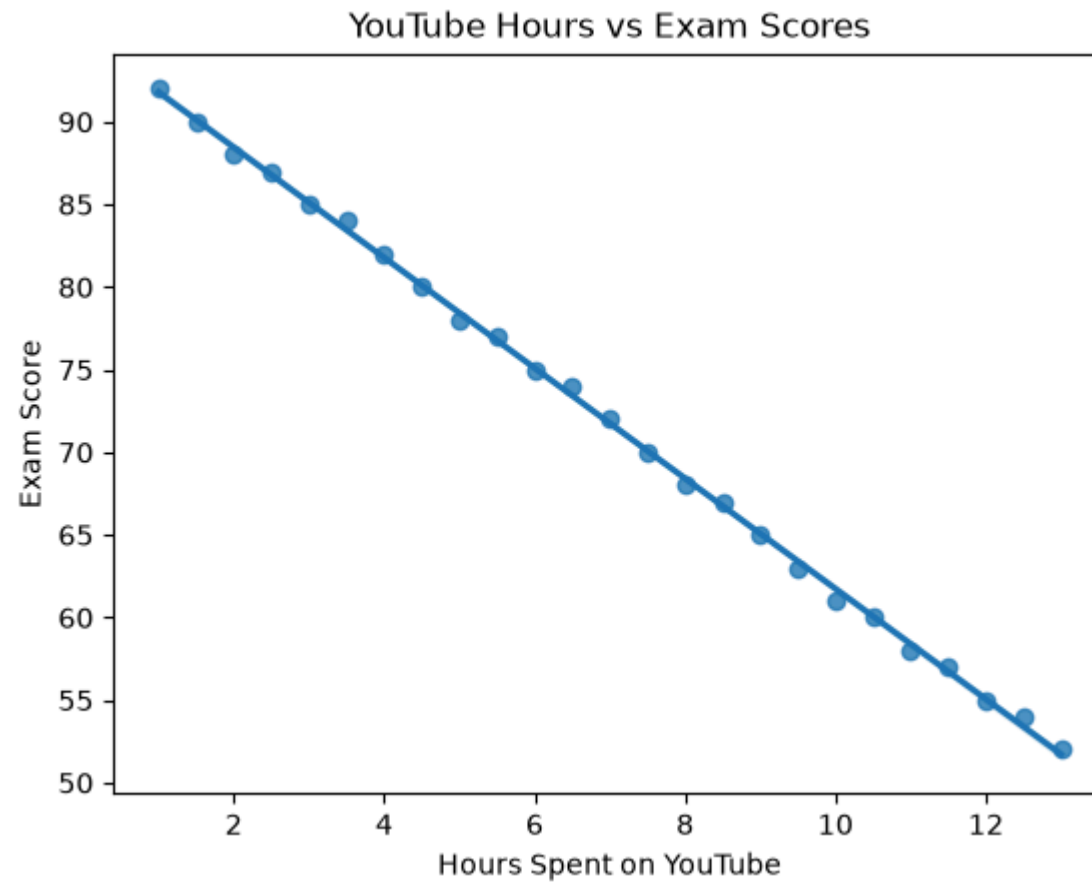
Out[32]:

	student	youtube_hours	exam_score
0	Student_1	1.0	92
1	Student_2	2.0	88
2	Student_3	1.5	90
3	Student_4	3.0	85
4	Student_5	2.5	87
5	Student_6	4.0	82
6	Student_7	3.5	84
7	Student_8	5.0	78
8	Student_9	4.5	80
9	Student_10	6.0	75
10	Student_11	5.5	77
11	Student_12	7.0	72
12	Student_13	6.5	74
13	Student_14	8.0	68
14	Student_15	7.5	70
15	Student_16	9.0	65
16	Student_17	8.5	67
17	Student_18	10.0	61
18	Student_19	9.5	63
19	Student_20	11.0	58
20	Student_21	10.5	60
21	Student_22	12.0	55

	student	youtube_hours	exam_score
22	Student_23	11.5	57
23	Student_24	13.0	52
24	Student_25	12.5	54

```
In [33]: sns.regplot(x="youtube_hours", y="exam_score", data = df)
plt.title("YouTube Hours vs Exam Scores")
plt.xlabel("Hours Spent on YouTube")
plt.ylabel("Exam Score")
```

```
Out[33]: Text(0, 0.5, 'Exam Score')
```



#Conclusion The regression plot shows a negative trend. In this sample dataset, as the number of hours spent on YouTube increases, exam scores generally decrease.